

Task_5-Microservices_with_Consul

Спрещу запускаємо consul.

```
sviatoslav:hw1$ consul agent -dev
==> Starting Consul agent...
    Version: '1.20.5'
    Build Date: '2025-03-11 10:16:18 +0000 UTC'
    Node ID: '44169901-afc2-a609-647e-9572a418e293'
    Node name: 'sviatoslav-IdeaPad-5-Pro-14ACN6'
    Datacenter: 'dc1' (Segment: '<all>')
    Server: true (Bootstrap: false)
    Client Addr: [127.0.0.1] (HTTP: 8500, HTTPS: -1, gRPC: 8502, gRPC-TLS: 8503, DNS: 8600)
    Cluster Addr: 127.0.0.1 (LAN: 8301, WAN: 8302)
    Gossip Encryption: false
    Auto-Encrypt-TLS: false
    ACL Enabled: false
    Reporting Enabled: false
    ACL Default Policy: allow
    HTTPS TLS: Verify Incoming: false, Verify Outgoing: false, Min Version: TLSv1_2
    gRPC TLS: Verify Incoming: false, Min Version: TLSv1_2
    Internal RPC TLS: Verify Incoming: false, Verify Outgoing: false (Verify Hostname: false), Min Version: TLSv1_2

==> Log data will now stream in as it occurs:

2025-04-27T23:02:40.328+0300 [DEBUG] agent.grpc.balancer: switching server: target=consul://dc1.44169901-afc2-a609-647e-9572a418e293/server.dc1 from=<none> to=<none>
2025-04-27T23:02:40.333+0300 [INFO] agent.server.raft: initial configuration: index=1 servers="[{Suffrage:Voter ID:44169901-afc2-a609-647e-9572a418e293 Address:127.0.0.1:8300}]"
```

Наступним кроком треба в consul записати значення, які будуть зчитуватися logging-service і messaging-service.

```
consul_setup.py > ...
1  import consul
2
3  c = consul.Consul()
4
5  c.kv.put('hazelcast/logging/cluster_name', 'logging-cluster')
6  c.kv.put('hazelcast/logging/members', '127.0.0.1:5701,127.0.0.1:5702,127.0.0.1:5703')
7  c.kv.put('messaging/queue/cluster_name', 'messaging-cluster')
8  c.kv.put('messaging/queue/members', '127.0.0.1:5801,127.0.0.1:5802')
9  c.kv.put('messaging/queue/queue_name', 'messages-queue')
```

← → ↻ localhost:8500/ui/dc1/kv 80% ☆



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⌵

⌵

dc1

Overview

Services

Nodes

Key/Value

Intentions

Access Controls

Tokens

Policies

Roles

Auth Methods

Organization

Peers

Key / Value

Type ▾

Name
hazelcast
messaging

← → ↻ localhost:8500/ui/dc1/kv/messaging/queue/members/edit 80% ☆



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dc1

Overview

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< Key / Values / messaging / queue

members

Value

JSON

⌵

⌵

```
1 127.0.0.1:5801,127.0.0.1:5802
```

Save

Cancel

Delete

Далі можна все запустити `./run_all.sh` і дивитися на логи.

```
2025-04-27T23:06:06.775+0300 [DEBUG] agent: Check in sync: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8002
2025-04-27T23:06:06.775+0300 [DEBUG] agent: Check in sync: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8003
2025-04-27T23:06:06.775+0300 [DEBUG] agent: Check in sync: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8001
2025-04-27T23:06:09.853+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8001 status=passing
2025-04-27T23:06:11.278+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8002 status=passing
2025-04-27T23:06:12.147+0300 [DEBUG] agent: Check status updated: check=service:messages-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8005 status=passing
2025-04-27T23:06:14.312+0300 [DEBUG] agent: Check status updated: check=service:messages-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8004 status=passing
2025-04-27T23:06:16.010+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8003 status=passing
2025-04-27T23:06:16.776+0300 [DEBUG] agent: Check status updated: check=service:facade-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8000 status=passing
2025-04-27T23:06:19.855+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8001 status=passing
2025-04-27T23:06:21.280+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8002 status=passing
2025-04-27T23:06:22.149+0300 [DEBUG] agent: Check status updated: check=service:messages-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8005 status=passing
```

```
INFO: 127.0.0.1:39088 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:39122 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:55312 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:53296 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:38142 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:43566 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:46576 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:51932 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:45776 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:41738 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:55692 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:51764 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:58054 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:36802 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:44832 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:53824 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:35434 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:44296 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:33506 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:56734 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:33938 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:38400 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:38704 - "GET /health HTTP/1.1" 200 OK
INFO: 127.0.0.1:39706 - "GET /health HTTP/1.1" 200 OK
```

Також, коли запускаємо всі компоненти, то вони записуються в consul.

← → ↺

localhost:8500/ui/dc1/services

80% ☆

⚙️

dc1

Overview

Services

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Key/Value

Intentions

Access Controls

Tokens

Policies

Roles

Auth Methods

Organization

Peers

Services 4 total

🔍 Search

Search Across ▾

Health Status ▾

Service Type ▾

✓ consul

1 instance

✓ facade-service

1 instance

✓ messages-service

2 instances

✓ logging-service

3 instances

Наступним кроком запишемо повідомлення від 1 до 5.

← → ↺

127.0.0.1:8000/docs#/default/create_message__post

90% ☆

Name

Description

msg ★ required

string

(query)

1

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/?msg=1' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

```
http://127.0.0.1:8000/?msg=1
```

Server response

Code	Details
200	Response body <pre>{ "status": "message created" }</pre>

Після цього спробую отримати результат. Все є.

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body <pre>{ "status": "ok", "logging_service_messages": "4;1;2;5;3", "messages_service_messages": "1;5" }</pre> Response headers <pre>content-length: 88</pre>

Далі виконаю "Продемонструвати, що у випадку відключення екземпляру певного мікросервісу, це буде відображатись у Consul (відключений екземпляр сервісу змінить статус) , а виклики будуть перенаправлятись до інших працюючих екземплярів."

Перед вимкненням одного із logging-services перевіримо їхні статуси.

← → ↺ localhost:8500/ui/dc1/services/logging-service/instances 80% ☆

dc1

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< All Services

logging-service

Instances Intentions Routing Tags

Q Search Search Across Health Status

logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8001	✓ All service checks passing	✓ All node checks passing	sviatslav-IdeaPad-5-Pro-14ACN6	localhost:8001
logging-service-sviatslav-IdeaPad-5-Pro-14ACN6-8002	✓ All service checks passing	✓ All node checks passing	sviatslav-IdeaPad-5-Pro-14ACN6	localhost:8002
logging-service-sviatslav-IdeaPad-5-Pro-14ACN6-8003	✓ All service checks passing	✓ All node checks passing	sviatslav-IdeaPad-5-Pro-14ACN6	localhost:8003

Вбиваємо один із logging-services.

```
sviatslav@sviatslav-IdeaPad-5-Pro-14ACN6: ... x sviatslav@sviatslav-IdeaPad-5-Pro-14ACN6: ... x sviatslav@sviatslav-IdeaPad-5-Pro-14ACN6: ...  
sviatslav:hw1$ lsof -i :8001  
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME  
python3 21457 sviatslav 18u IPv4 98805 0t0 TCP *:8001 (LISTEN)  
sviatslav:hw1$ kill -9 21457  
sviatslav:hw1$
```

Бачимо, що тепер статус одного з сервісів critical.

```
sviatslav@sviatslav-IdeaPad-5-Pro-14ACN6: ... x sviatslav@sviatslav-IdeaPad-5-Pro-14ACN6: ... x sviatslav@sviatslav-IdeaPad-5-Pro-14ACN6: ... x  
2025-04-27T23:10:39.903+0300 [WARN] agent: Check is now critical: check=service:logging-service-sviatslav-IdeaPad-5-Pro-14ACN6-8001  
2025-04-27T23:10:41.334+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatslav-IdeaPad-5-Pro-14ACN6-8002 status=passing  
2025-04-27T23:10:42.204+0300 [DEBUG] agent: Check status updated: check=service:messages-service-sviatslav-IdeaPad-5-Pro-14ACN6-8005 status=passing  
2025-04-27T23:10:44.366+0300 [DEBUG] agent: Check status updated: check=service:messages-service-sviatslav-IdeaPad-5-Pro-14ACN6-8004 status=passing  
2025-04-27T23:10:46.070+0300 [DEBUG] agent: Check status updated: check=service:logging-service-sviatslav-IdeaPad-5-Pro-14ACN6-8003 status=passing  
2025-04-27T23:10:46.835+0300 [DEBUG] agent: Check status updated: check=service:facade-service-sviatslav-IdeaPad-5-Pro-14ACN6-8000 status=passing  
2025-04-27T23:10:49.904+0300 [WARN] agent: Check is now critical: check=service:logging-service-sviatslav-IdeaPad-5-Pro-14ACN6-8001
```

← → ↻ localhost:8500/ui/dc1/services/logging-service/instances 80% ☆

dc1

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< All Services

logging-service

Instances Intentions Routing Tags

🔍 Search Search Across Health Status

logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8001
❌ All service checks failing ✅ All node checks passing ➡ sviatoslav-IdeaPad-5-Pro-14ACN6 🌐 localhost:8001
logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8002
✅ All service checks passing ✅ All node checks passing ➡ sviatoslav-IdeaPad-5-Pro-14ACN6 🌐 localhost:8002
logging-service-sviatoslav-IdeaPad-5-Pro-14ACN6-8003
✅ All service checks passing ✅ All node checks passing ➡ sviatoslav-IdeaPad-5-Pro-14ACN6 🌐 localhost:8003

Попри це можемо далі працювати із сервісом.

← → ↻ 127.0.0.1:8000/docs#/default/get_messages__get 90% ☆

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/
```

Server response

Code	Details
200	Response body <pre>{ "status": "ok", "logging_service_messages": "4;1;2;5;3", "messages_service_messages": "2" }</pre> Response headers <pre>content-length: 86 content-type: application/json date: Sun, 27 Apr 2025 20:11:37 GMT</pre>